The usage of AI

Ai Acknowledgement

AI tools were used to assist with coding, visualisation, and minor language improvements in this assignment.

1. Assisted with R coding (e.g., data cleaning, and plotting)
2. Supported ggplot visualisation (e.g., colours, scaling, regression plots)
3. Helped interpret and describe figures (e.g., boxplots, regression)
4. Assisted with Quarto formatting (e.g., inserting images, rendering PDF)
5. Corrected the English.
   The screenshots are shown below

Ai Screenshots

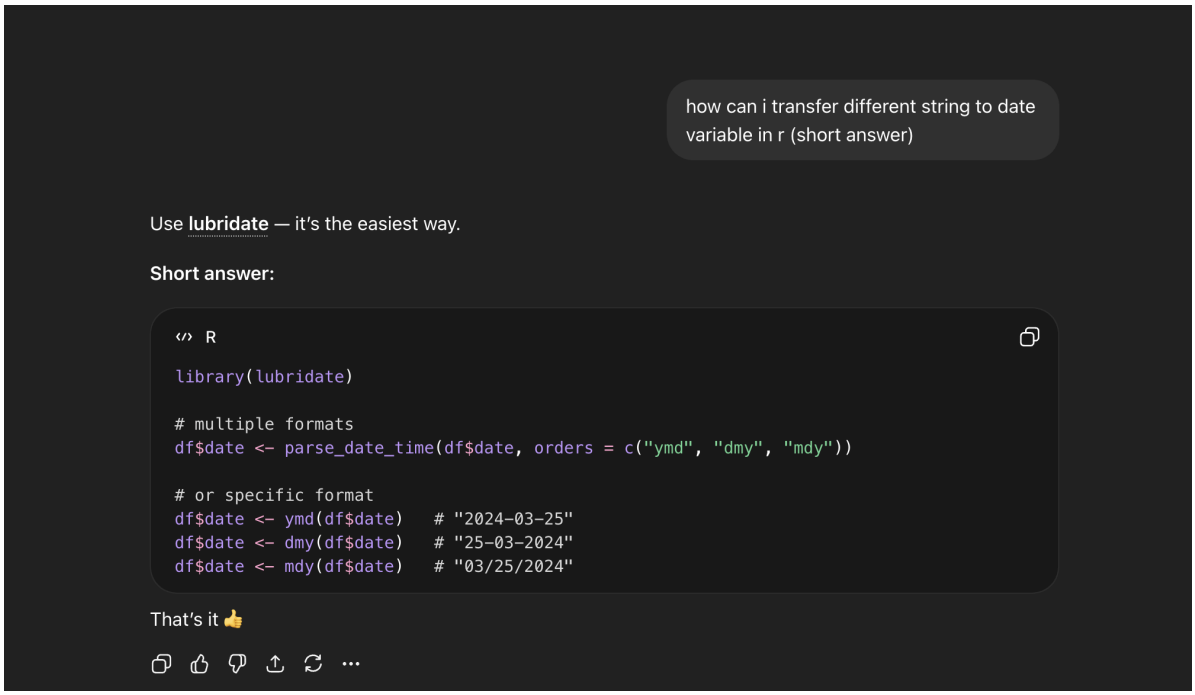


Figure 1: R code for transfering to date

how can i color different group when i use ggplot

Use the `color` or `fill` aesthetic in **ggplot2**.

Short answer:

1. For points / lines → use `color`

```r
ggplot(data, aes(x = x_var, y = y_var, color = group_var)) +
  geom_point()
```

2. For bars / boxplots → use `fill`

```r
ggplot(data, aes(x = group_var, y = value, fill = group_var)) +
  geom_boxplot()
```

3. Custom colors (optional)

```r
+ scale_fill_manual(values = c("red", "blue", "green"))
```

👉 Key idea:

- `color` = outline / points / lines
- `fill` = inside of shapes (bars, boxes)

If you want, I can fix your exact plot so it looks nicer 👍

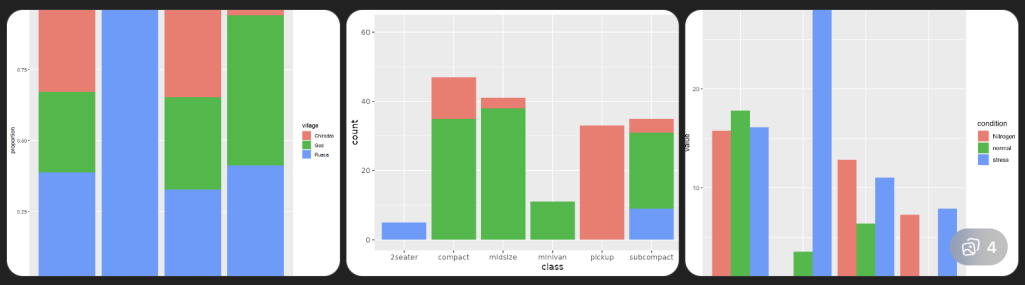
Figure 2: R code for group colors in plot

what are the types of plot in ggpolt

Here are the **main types of plots** in ggplot2 (short + useful for your assignment):

📊 Common plot types

1. Bar plot (counts / categories)




```R
ggplot(data, aes(x = category)) +
  geom_bar()
```

👉 Use for: number of observations per group (e.g. segment count)

Figure 3: Types of ggplot

Figure 4: Types of ggplot

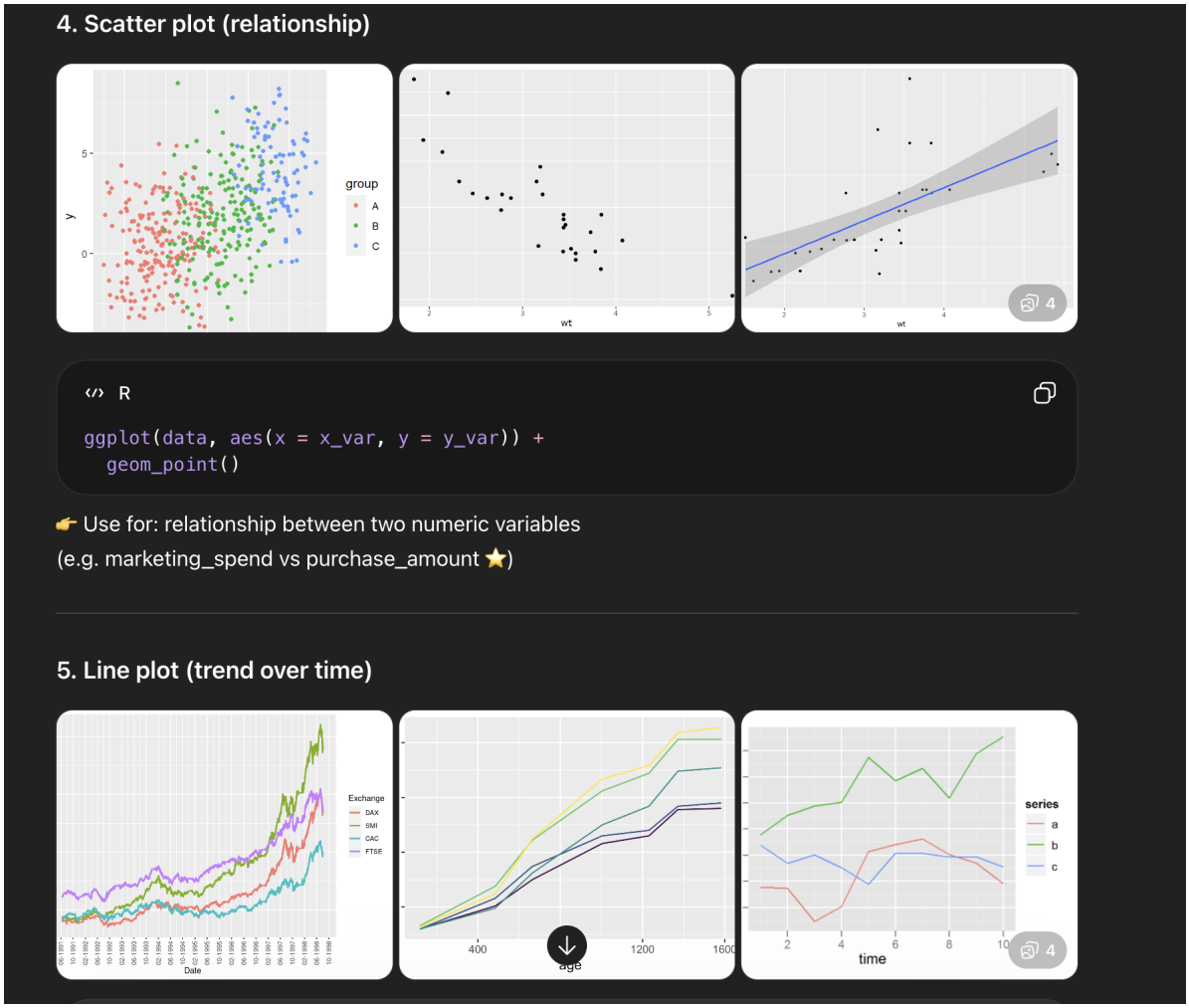


Figure 5: Types of ggplot

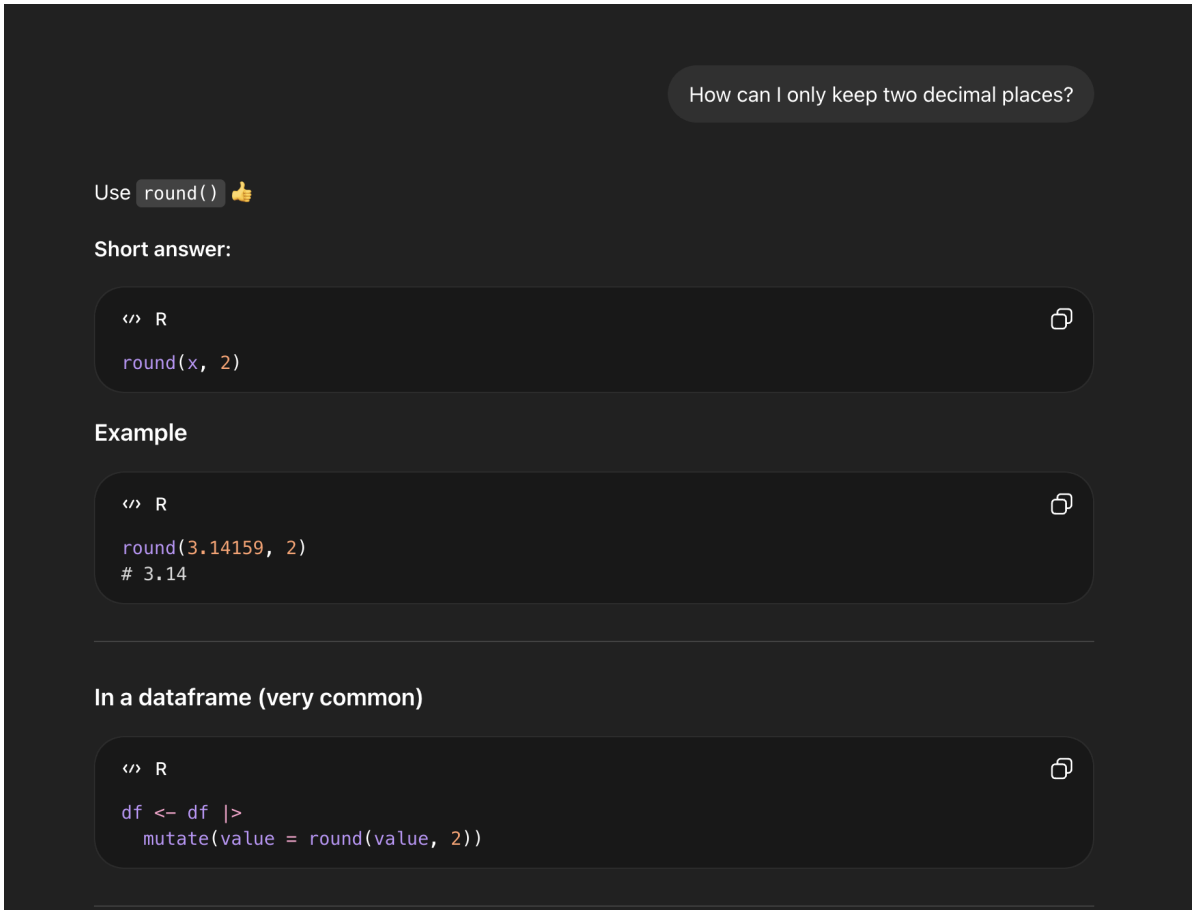


Figure 6: R code for save decimal place

how can i change the scale of y-axis(zoom in only)

Use `coord_cartesian()` 👍 (this is the correct way to *zoom in only* in **ggplot2**)

✅ Zoom in (recommended)

```r
ggplot(data, aes(x, y)) +
  geom_point() +
  coord_cartesian(ylim = c(0, 100))
```

👉 This:

- **zooms the view only**
- **keeps all data** (important for boxplots, stats, etc.)

❌ Don't use this for zoom

```r
ylim(0, 100)
```

👉 This **removes data outside the range** (can mess up your plot)

Figure 7: R code for zoom y-axis

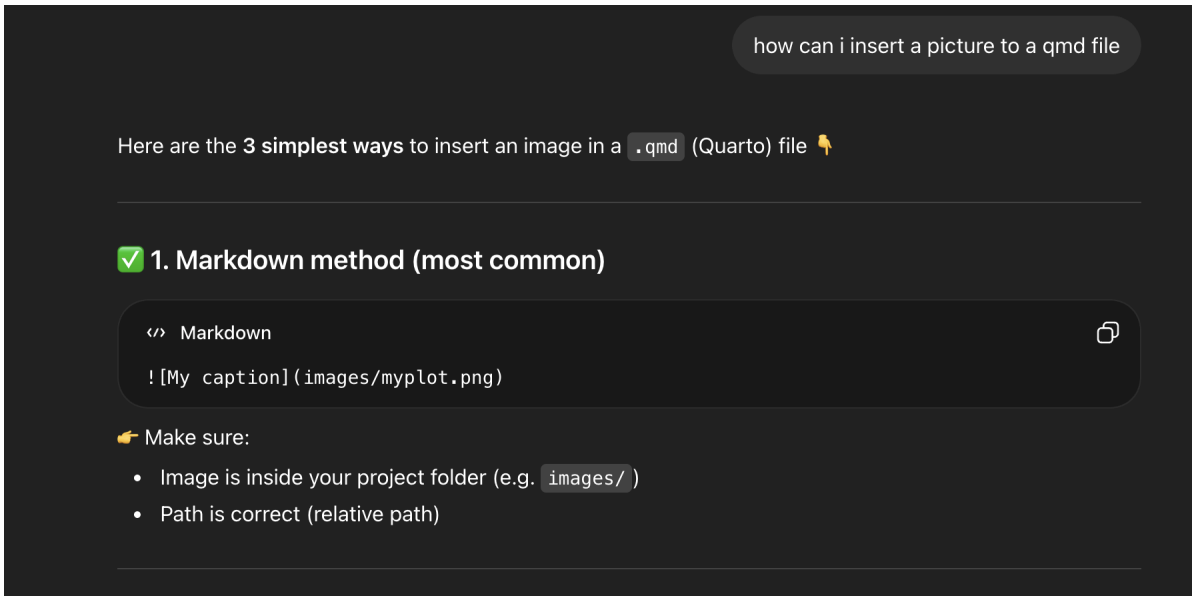


Figure 8: R code for insert picture

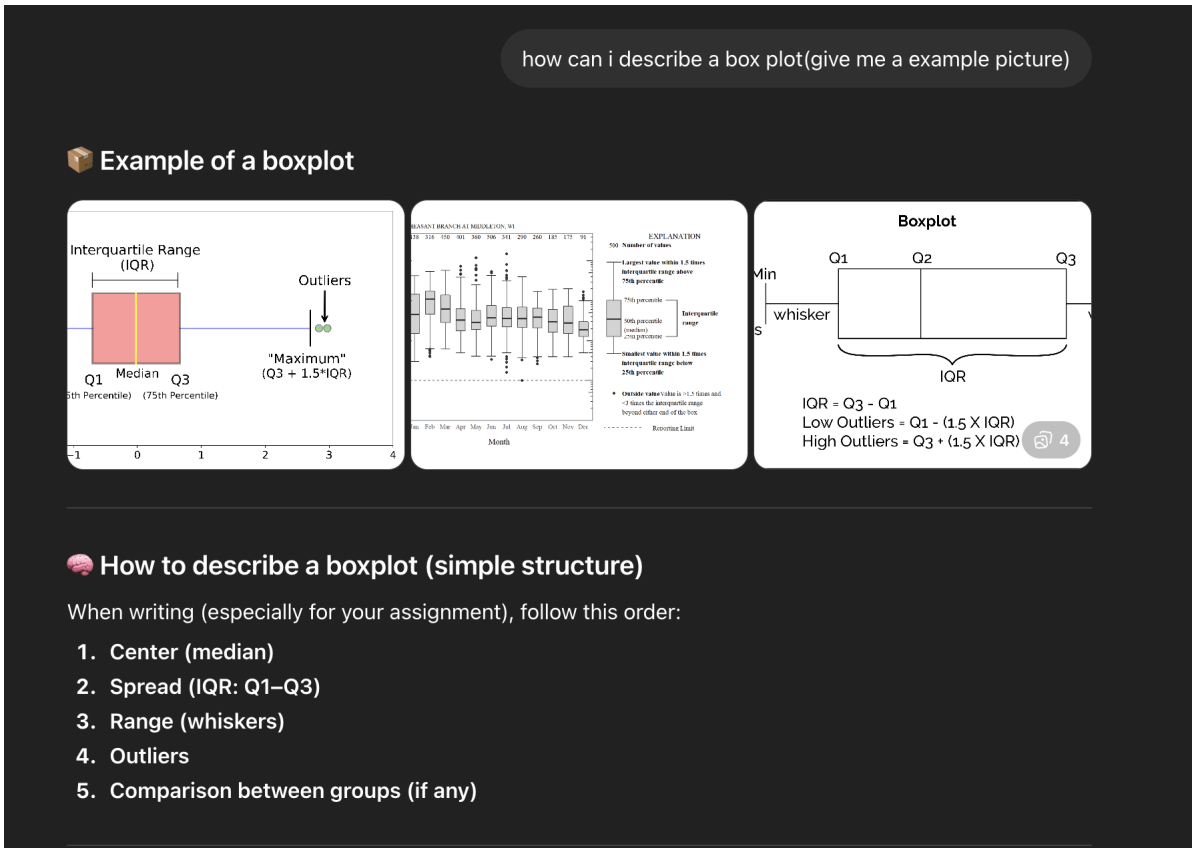


Figure 9: help to describe boxplot

how to plot a linear regression

Here's the **quick way** to plot a linear regression in **ggplot2** 👇

✅ 1. Basic linear regression plot

R

```r
ggplot(data, aes(x = x_var, y = y_var)) +
  geom_point() +
  geom_smooth(method = "lm")
```

👉 This gives:

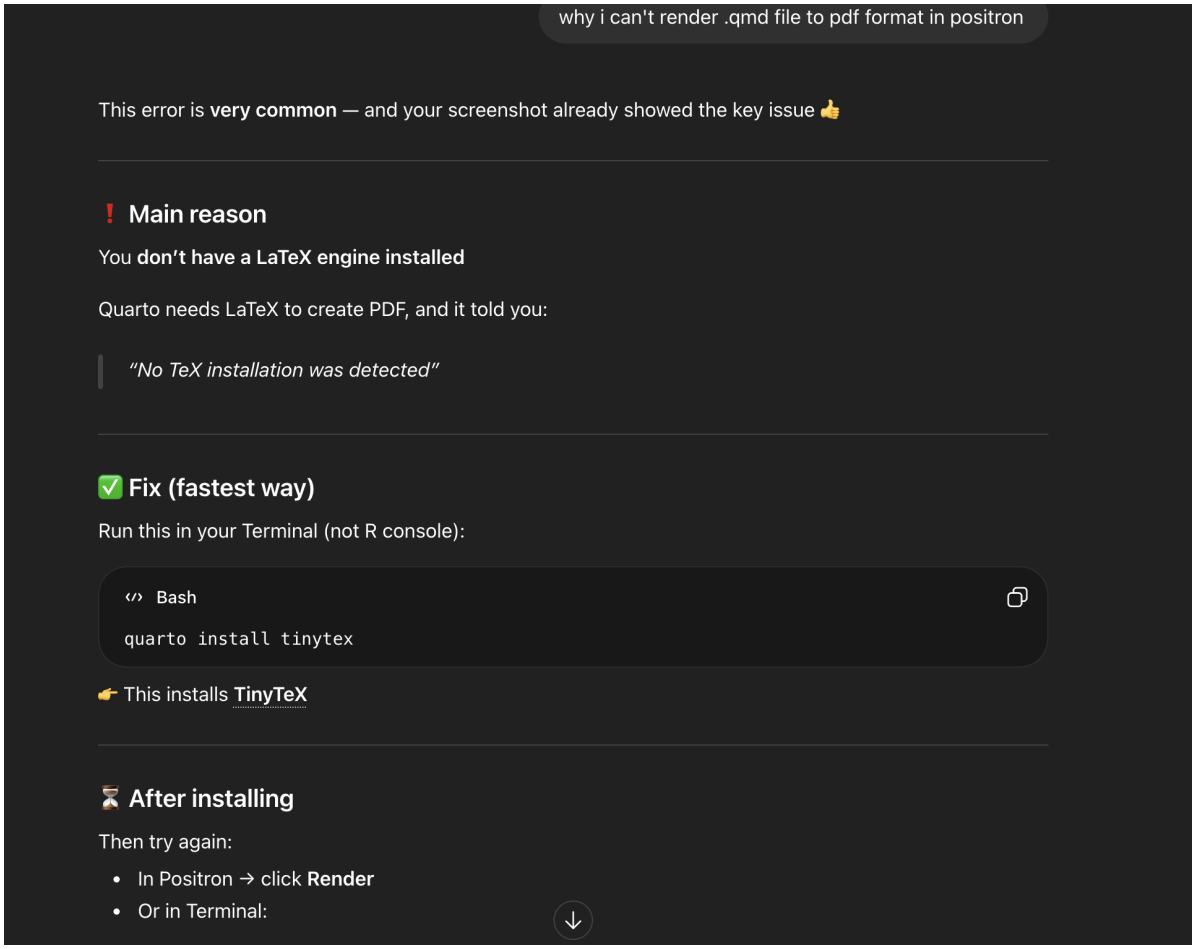
- scatter plot
- • regression line (automatically)

✅ 2. Remove shaded confidence interval (cleaner)

R

```r
ggplot(data, aes(x = x_var, y = y_var)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

Figure 10: R code for plot linear regression



why i can't render .qmd file to pdf format in positron

This error is **very common** — and your screenshot already showed the key issue 👍

❗ Main reason

You **don't have a LaTeX engine installed**

Quarto needs LaTeX to create PDF, and it told you:

> *"No TeX installation was detected"*

✅ Fix (fastest way)

Run this in your Terminal (not R console):

```bash
quarto install tinytex
```

👉 This installs **TinyTeX**

⏳ After installing

Then try again:

- In Positron → click **Render**
- Or in Terminal:

Figure 11: help to render to pdf

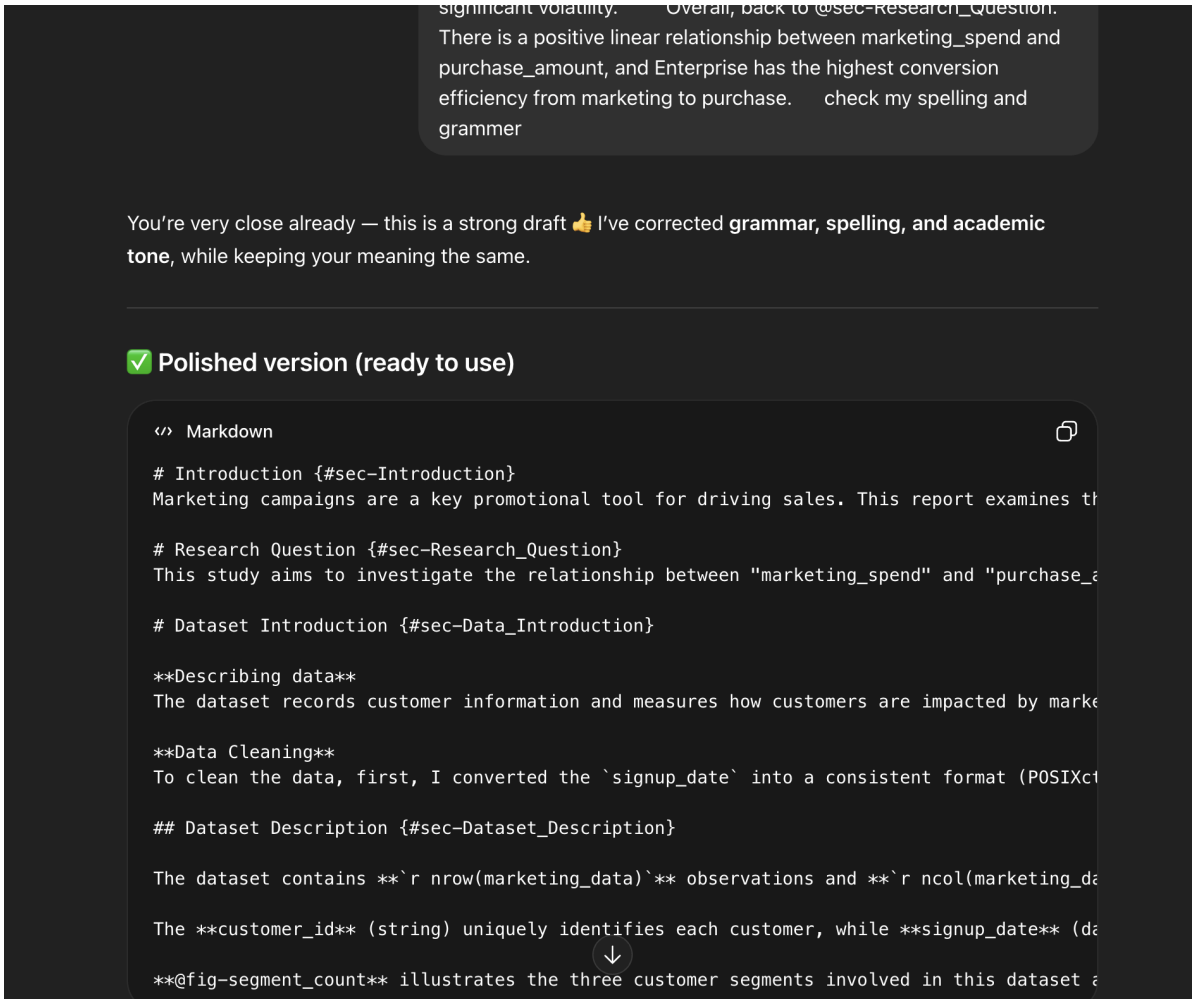


Figure 12: help to check the English